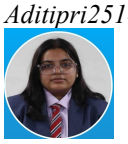
PROJECT AND TEAM INFORMATION

Project Title

Home Energy Analytics and Budget Advisor

Team Name: *Team #* Team ID: DSCPP-III-2026-T127	*Tech Titans*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):* *ruhanianeja29@gmail.com* 	*Aneja, Ruhani – 2510370225*
Team member 2 *(Last Name, name: student ID: email, picture):* *suryanshtayal2007@gmail.com* 	*Tayal, Suryansh – 2510370336*
Team member 3 *(Last Name, name: student ID: email, picture):* *Aditipri2514@gmail.com* 	*Gupta, Aditi – 2510380082*

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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

Households rarely know which devices drive their power bill. Our project addresses this with a command-line C++ tool that reads a dataset of 100 made-up appliance entries and works out how much electricity each device likely uses and what it costs. Every row of the CSV describes one appliance through its name, category, rated wattage, typical hours of use per day, a priority level and a status flag.

After loading the rows, the program turns wattage and usage hours into energy figures and money estimates, then compares the totals against a budget so the user gets practical suggestions on where to cut back. The work draws on file handling, string manipulation, and core data structures and algorithms for storing, looking up, ordering and grouping records.

All figures come from the simulated file. The tool does not connect to a smart meter and does not switch any real device on or off.

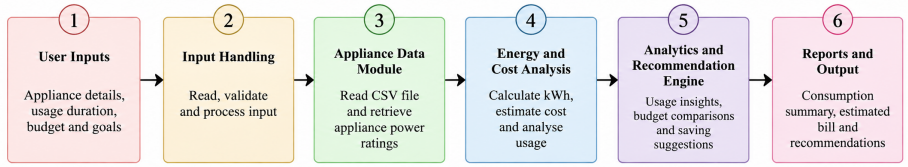
Updated Project Approach and Architecture

The design is built around one input file, `electricity_management_data.csv`, and one source file, `main.cpp`. At start-up each line is split into fields, checked, and kept in memory as an appliance record. The user then picks an option from a menu, and the program runs the matching calculation over those stored records and prints the outcome.

Implementation relies on the C++ standard library: `fstream` to open the file, `sstream` and `string` to cut each line into fields, and `vector` and `map` to hold and group the records. Lookup and ordering routines operate on the same in-memory collection.

Data flow: CSV file → parsing and validation → appliance records in memory → energy and cost computation → budget and category summaries → output on screen. Everything runs locally; no network communication or hardware sensors are involved.

System Architecture Diagram



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Tasks Completed

Task Completed	Team Member
Project topic and workflow planning	Team
Appliance data structure and CSV-based approach outlined	Suryansh Tayal / team
C++ program structure and data-processing work	Suryansh Tayal (technical work)
Report preparation and coordination	Ruhani Aneja
Initial idea exploration	Aditi Gupta

Challenges/Roadblocks

Unreliable input rows: A blank or wrongly formatted field can break parsing or produce wrong totals. We still need to feed the program empty files and damaged rows to see how it reacts.

Locating the data file: The program finds the CSV only if it sits next to the executable or if the path is written into the source. We must confirm this before the demonstration.

Realism of results: Because the dataset is invented, the computed bills will differ from what a real meter would show.

Meaning of the status column: This field is part of the simulation, not a live reading, so it must not influence the energy and cost formulas.

Missing test records: No test cases, outputs or screenshots have been recorded yet. These must be collected before the final submission.

Tasks Pending

Task Pending	Team Member (to complete the task)
Test valid, missing and malformed CSV input	Suryansh Tayal / team
Verify energy and cost calculations against manual examples	Suryansh Tayal / team
Verify search, sorting and category totals where implemented	Suryansh Tayal / team
Capture real execution screenshots and sample outputs	Ruhani Aneja / team
Final report review and formatting	Ruhani Aneja
Review project ideas and final presentation	Aditi Gupta / team

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Project Outcome/Deliverables

The end product is a C++ program that takes the simulated appliance file and reports estimated power consumption and bill amounts, together with budget guidance for the household.

Depending on what the current build of `main.cpp` supports, the user can look up a particular appliance, view the list in a chosen order, see totals per category, and read suggestions on staying within budget.

Alongside the program, the submission contains the source file `main.cpp`, the dataset `electricity_management_data.csv`, and this Phase-II report. Together they show how file handling, string processing and data structures and algorithms can be combined to solve an everyday problem. The tool is an offline simulation; it is neither a smart-meter interface nor a machine-learning system.

Progress Overview

So far the team has fixed the topic, the goal and the CSV-driven design, and is writing the report around the C++ program. Ruhani Aneja handles documentation, Suryansh Tayal handles the coding, and Aditi Gupta contributed early ideas.

Testing has not been completed. Before collecting evidence, the team will check that input loading, the cost formulas and the analysis options behave correctly. A feature will be listed as finished only after it has been confirmed in the current version of the program.

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Codebase Information

Repository: `https://github.com/ruhanianeja29/Electricity-consumption-analyzer`
Main source file: `main.cpp`
Dataset: `electricity_management_data.csv`
Language: C++
Input method: Local CSV file.
Build command (if using g++): `g++ main.cpp -o electricity`
Branch and important commits: To be filled after checking the current repository.

Testing and Validation Status

Test Type	Status	Notes
CSV dataset loading	Partially tested / verify	Confirm expected records are loaded.
Missing or empty CSV	Pending verification	Check error handling.
Malformed CSV rows	Pending verification	Check handling of invalid fields.
Energy and cost calculation	Verify with sample	Compare output with manual calculation.
Appliance search and sorting	Pending verification	Test known/missing names and ordering if supported.
Category totals and budget checks	Pending verification	Compare displayed values with manual results if supported.
Report/output display	Partially tested / verify	Save a genuine sample output.

Deliverables Progress

Deliverable	Current Status
C++ source code (`main.cpp`)	Available / verify current build
CSV appliance dataset	Available / verify contents
Energy and cost analysis features	In progress; validate outputs
Project report	In progress / updated draft
Testing evidence and screenshots	Pending / partially complete
GitHub repository information	Link provided; branch to verify
Final Phase-II submission	Pending final review

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